

Spatiotemporal graph convolutional network using sparse monitoring data for accurate water-level reconstruction in urban drainage systems

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ABSTRACT

A comprehensive monitoring of urban drainage network (UDN) is essential for maintenance, management, and sustainable urban development. However, limited sensor deployment hinders the acquisition of sufficient information. Conventional deep learning methodologies can predict and correct monitored data but struggle with unobserved data. Hydraulic models can simulate behaviors but face data collection challenges and low real-time performance. To address these issues, a novel spatiotemporal graph convolutional network (STGCN) model, based on graph neural networks, is proposed to reconstruct a real-time information system for UDNs. By extracting fundamental elements from limited monitoring data and UDN topology, the STGCN model effectively reconstructed unmonitored node data. The experimental results showed that the training efficiency and reconstruction accuracy of the model could be optimized by reducing the spatial data dimensionality to 0.6, adopting a passive-masked training strategy with a ratio of 4:3 for model-training sensors to loss-calculation sensors, and using a historical data input length of 3 h. This approach allowed for the reconstruction of water levels for 527 unmonitored nodes using only seven monitoring nodes, with a median mean absolute error of 0.038 m and an accuracy of 71.3 %. These results demonstrate that the STGCN model can accurately reconstruct unmonitored node data using low monitoring-node density and basic network topology, offering a practical solution to data-driven challenges in intelligent UDNs. The source code is available at https://github.com/holylove9412/UDNs_STGCN_model.

1. Introduction

Urban drainage networks (UDNs) are critical components of contemporary urban infrastructure, with their operational efficiency significantly impacting economic development and societal sustainability (Zinatloo-Ajabshir et al., 2024). The evolution of UDNs hinges on intelligent and efficient network management facilitated by the collection and analysis of substantial amounts of operational data. However, challenges include delays in digitizing information, complexities of monitoring environments, operational and maintenance difficulties, and substantial management-related costs (Olsson, 2021; Radhakrishnan & Wu, 2018; Yang et al., 2020). These challenges lead to sparse monitoring networks and compromised data integrity. In such scenarios, inferring unmonitored information within UDNs from limited monitoring-node

density becomes crucial for improving the operational efficiency and resilience of urban infrastructure, as well as guiding UDNs toward intelligent development.

Accurate monitoring data are crucial for the intelligent and efficient management of UDNs. Thereby, inconsistencies or deficiencies in these data must be identified and rectified to ensure efficient operation. Previous studies have addressed this issue by developing methods for reconstructing missing or abnormal sensor data. Cugueró-Escofet et al. (2016) introduced a method for the reconstruction of missing or abnormal sensor data by integrating spatial and time-series models. He et al. (2023) utilized a combination of 1D convolution and gated recurrent unit models to effectively capture both the long-term and local hydraulic characteristics across several drainage networks, facilitating the rectification of missing water-level data. Choi et al. (2023)

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implemented a context-aware voting ensemble model, enabling the imputation of missing water-quality data across various scenarios via dynamic selection. However, sparse monitoring coverage remains a challenge for accurate decision support in intelligent drainage systems, even with recovered detection data. Hence, reconstructing comprehensive system information from a limited number of monitoring nodes is crucial for effective operation and maintenance.

High-precision mechanism models, established by leveraging accurate geographic information system (GIS) for water networks and operation monitoring data related to the flow and level of water, facilitate Digital Twin realization and accurate representation of actual network systems (Farina et al., 2023; Shiu et al., 2024; Zanfei et al., 2023; L. Zhang et al., 2023). Simulating water network operations under various conditions enables the reconstruction of corresponding water network behaviors, effectively addressing challenges in the optimization of water distribution networks (Sitzenfrei et al., 2020), detection and localization of pipe bursts and leaks (Zaman et al., 2020), flood control and drainage optimization (Haghighatafshar et al., 2019; Teng et al., 2017), as well as pump and gate control optimization (Zhang et al., 2023). However, technological limitations have resulted in UDNs facing issues such as inadequate foundational data, compromised pipe function attributes, and limited availability of operation monitoring data, substantially hindering model construction and calibration processes (Fu et al., 2022; Hutton et al., 2014; Li et al., 2024; Preis et al., 2011). Moreover, the high computational resource demand of this approach makes it challenging to achieve a real-time monitoring of network operation states (Zhou et al., 2022).

The advancement of scientific and technological innovation, particularly the improvement of computational capacity, has enabled the efficient processing of large-scale datasets. This development has significantly advanced the field of data mining based on machine learning, which has produced remarkable results in processing structured data (such as images and speech) and unstructured data (such as text) in the Euclidean space (Fu et al., 2022; Fu et al., 2022; Shinde & Shah, 2018). Data in the Euclidean space have regularity and fixed proximity in the physical space, enabling conventional deep learning models, such as convolutional neural networks (CNNs), to effectively capture local features and accommodate the translational invariance in data through convolution operations (Wu et al., 2021). However, with continued research on complex data structures, there are increasingly more applications involving non-Euclidean data structures, such as social networks (Zheng et al., 2024), chemical molecular structures (Jiang et al., 2021), and protein interaction networks (Yuan et al., 2021). These data are essentially node-and-edge graph structures with significant irregularities, which can represent the intricate relationships between entities and interactions. Conventional deep learning models encounter significant difficulties with non-Euclidean data due to their inability to interpret irregular topological structures and their emphasis on the concept of scale and distance (Wu et al., 2021). Therefore, graph neural networks (GNNs) have been specifically developed to address graph-structured data (Kipf & Welling, 2017). Unlike conventional deep learning models, a GNN can directly learn from a graph structure, effectively capturing complex relationships and feature information between nodes through information propagation and aggregation mechanisms (Zhou et al., 2020).

In recent years, the far-reaching impact of GNNs in many fields, such as transport (Li & Zhu, 2021), finance (Liu et al., 2022), composites (Zinatloo-Ajabshir et al., 2019), and biosciences (Lim et al., 2019), has promoted their application in water distribution systems. Wu et al. (2023) employed a GNN to translate prior hydraulic knowledge into machine learning methods for water demand forecasting. Buchhorn et al. (2023) proposed a graph deviation network for sensor anomaly detection in river network systems. Li et al. (2024) utilized a gated GNN to estimate water-quality data across all nodes in an entire water distribution system. Despite the successful application of GNNs to water distribution systems, UDNs pose unique and challenging problems, such

as complex and diverse hydraulic characteristics typically involving both gravity and pressure flows. Moreover, UDNs are influenced by multiple factors, including topography, precipitation, soil types, and human activities, resulting in highly complex, dynamic, and unstable operation monitoring data (Troutman et al., 2017; Zhang et al., 2018). Hence, it is necessary to integrate conventional deep learning models with GNNs to effectively handle multimodal data in drainage systems.

The backward informatization of UDNs has led to a prevalent issue: lack of basic data. This makes it extremely difficult to collect different types and quantities of spatial static data and temporal dynamic data. Existing studies often use simulated values of all nodes from hydraulic models to calculate the training losses; this approach is impractical in real-world scenarios for acquiring monitoring data for all nodes in a UDN (Li et al., 2024).

This study proposes a spatiotemporal graph convolutional network (STGCN) model based on the GNN for reconstructing water levels at unmonitored nodes in UDNs. By extracting the spatiotemporal feature information of UDNs from limited multimodal data sources, including spatial static data (network topology) and temporal dynamic data (rainfall data, water-level monitoring data), which are the most readily available in real-world scenarios, the STGCN model is expected to maximize the use of limited information to learn the hydraulic correlations between monitored and unmonitored nodes, thus accurately reconstructing the water levels at all unmonitored nodes within a UDN. This study focused on 1) utilizing dimensionality reduction on a normalized diffusion matrix ($\tilde{A}$) to enhance the model training efficiency, 2) adopting a novel masked training approach to enhance the transfer learning ability and flexibility of the model to optimize its generalizability for unmonitored nodes, 3) comparing and analyzing the impact of historical data from monitoring points of varying lengths on the model performance.

2. Methodology

In real-world UDNs, the installation of water-level monitoring sensors is often limited to a small number of nodes due to various internal and external factors such as cost constraints, environmental conditions, and technical challenges. This limitation leads to insufficient monitoring of the overall operational status of a drainage system. This study focused on reconstructing the water-level data at unmonitored nodes using data from a limited number of monitoring nodes to facilitate subsequent intelligent optimization and scheduling applications.

This study aimed to develop a water-level monitoring information system for the entire UDN by effectively extracting the spatiotemporal relationships between rainfall data, drainage network topology, and water-level monitoring data from limited nodes using the STGCN model. Fig. 1 shows the workflow of the STGCN-based water-level monitoring information system, where the details of each component are described in the subsequent sections. Section 2.1 provides detailed processing procedures for both spatial and temporal data. Section 2.2 introduces the overall framework of the STGCN model. Section 2.3 presents a flowchart of the model training strategy. Section 2.4 provides evaluation metrics for model training, validation, and testing.

2.1. Preprocessing of multimodal types of data

The multimodal types of data leveraged in this study primarily encompassed spatial data associated with the topology of the UDN as well as temporal data related to rainfall and water levels. The spatial static data were encoded as adjacency matrices, whereas the temporal dynamic data were represented as vectors.

The UDN comprised n nodes and m pipelines. To monitor the water-level fluctuations at the nodes, sensors were deployed at s ($s < n$) nodes. The topological structure of a UDN can be represented by the topology diagram $G = (V, E)$, where V denotes the node datasets in the system

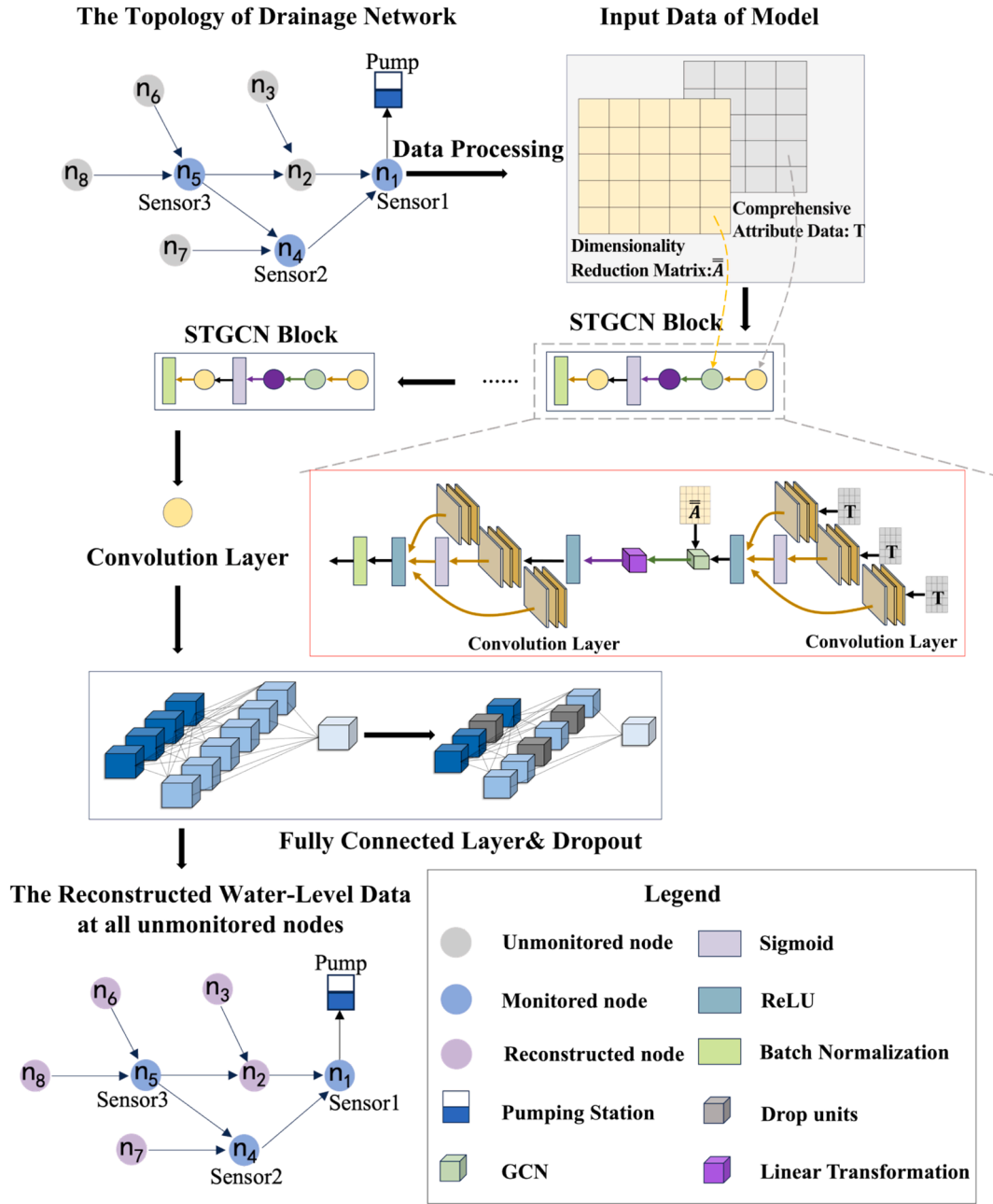


Fig. 1. The workflow of the STGCN-based water-level monitoring information system.

corresponding to drainage outlets, pumping station forebays, and inspection wells; E represents the set of edges, corresponding to pipe segments and pumps.

2.1.1. Spatial static data processing

Fig. 2 shows the processing of the topology (spatial static data) of the UDN. The adjacency matrix $A \in R^{n \times n}$, reflecting the node connection relationship in the graph structure, is constructed based on the topology diagram G . The connectivity between nodes i and j is represented by the matrix element A_{ij} , as expressed in Eq. (1). To enable each node to incorporate its own information while concurrently aggregating information from adjacent nodes, a self-connection (achieved by adding A to the identity matrix I) is introduced for every node, as expressed in Eq. (2). After the incorporation of self-connections, A is normalized to produce a normalized diffusion matrix $\bar{A}$, as expressed in Eq. (3). This process is aimed at ensuring uniform information dissemination

throughout the network so that all the nodes are endowed with comparable metric ranges in the initial state.

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & A_{ij} & \vdots \\ A_{n1} & \cdots & A_{nn} \end{bmatrix} \quad (1)$$

$$\bar{A} = A + I \quad (2)$$

$$\tilde{A} = D^{-\frac{1}{2}} \times \bar{A} \times D^{-\frac{1}{2}} \quad (3)$$

$$D_{ii} = \sum_j A_{ij} \quad (4)$$

where A_{ij} represents the components of the adjacency matrix A ; the subscripts i and j represent the nodes of the pipe segment, respectively; when A_{ij} is equal 1, it signifies that the flow direction is from i to j , and when it is equal to 0, the flow direction is not from i to j ; I represents the

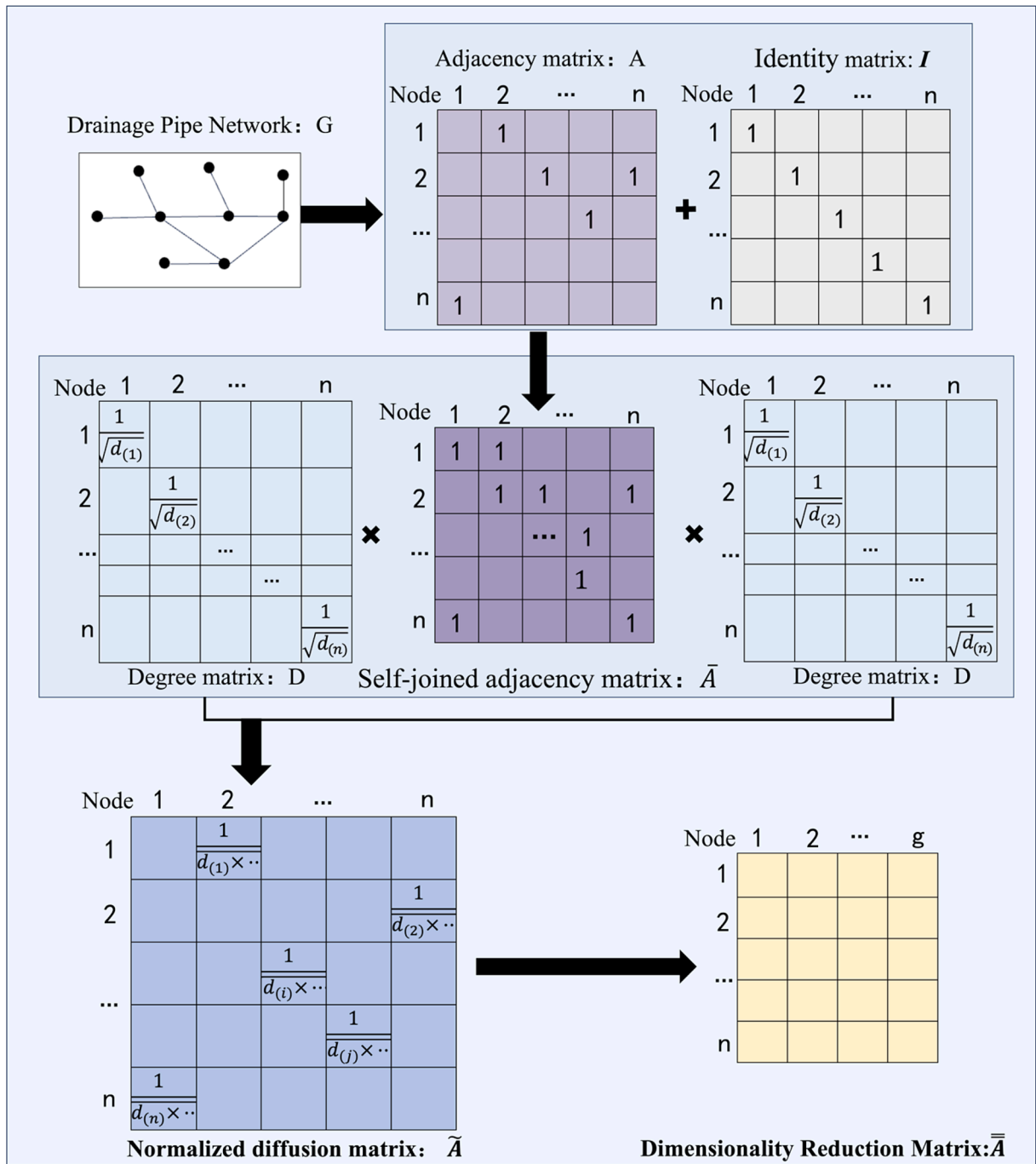


Fig. 2. The processing of spatial static data (UDN's topology).

identity matrix; D is a diagonal matrix, and the diagonal elements D_{ii} represent the degree of node i (i.e., the number of edges connected to node i , including self-connections) as expressed in Eq. (4).

The data structure of $\tilde{A}$ is sparse and high-dimensional, making it challenging for the model to extract useful information from it. This can lead to a low training efficiency and a highly complex training process. Efficient dimensionality-reduction techniques, aimed at eliminating noise and redundant information, can transform high-dimensional sparse data into low-dimensional dense data. This approach significantly enhances the training efficiency while also assisting the model in more accurately identifying critical features.

The dimensionality reduction processing of the high-dimensional sparse matrix $\tilde{A}$ was performed using singular value decomposition

(SVD). SVD employs g singular values with their corresponding left and right singular vectors to approximate the description of $\tilde{A} \in R^{n \times n}$, thus representing the main spatial connectivity features of the UDN with $\bar{\bar{A}} \in R^{n \times g}$ (Klema & Laub, 1980). By selecting an appropriate g value, the stability of model training and the accuracy of the output results could be enhanced by preserving the important properties of the data while minimizing the number of features. In this part, the principal component analysis was employed to identify the minimal dimension g , where the dataset could preserve a specified proportion of its effective information (Abdi & Williams, 2010).

2.1.2. Temporal dynamic data processing

Water-level information, which is an important indicator reflecting

the operational status of UDNs, served as the primary research variable in this study. Considering that precipitation conditions significantly influence water-level data, rainfall data were also considered. Fig. 3 shows the processing of these two temporal dynamic datasets. The water-level and rainfall data of node i in the UDN during a certain period t_k are denoted by $h_t^{level} \in R^{1 \times k}$ and $h_t^{rain} \in R^{1 \times k}$, as expressed in Eq. (5), respectively. In this study, a uniform rainfall was assumed, meaning that the rainfall input for each node was the same at any given time. For nodes lacking water-level/rainfall data, the corresponding entries were filled with 0. H_{level} and H_{rain} represent the water-level and rainfall data of the UDN, as expressed in Eq. (6). As expressed in Eq. (7), H_{level} and H_{rain} are concatenated along the depth channel to generate a comprehensive node attribute data T . Subsequently, the node attribute data are aggregated using a sliding window to create the model dataset (Chen et al., 2022). This process required specifying the window length and stride length.

$$h_t^{i_{type}} = [f_{t_0}^{(i)} \dots f_{t_{k-1}}^{(i)} f_{t_k}^{(i)}], i \in V \quad (5)$$

$$H_{type} = [h_t^{1_{type}}, h_t^{2_{type}} \dots h_t^{n_{type}}], type \in (level, rain), t = \{t_0 \dots t_k\} \quad (6)$$

$$T = \begin{cases} H_{rain} \in R^{n \times k} \\ H_{level} \in R^{n \times k}, T \in R^{n \times k \times 2} \end{cases} \quad (7)$$

where $f_t^{(i)}$ is the water-level value or rainfall intensity of node i at time point t ; i belongs to the node dataset V ; $h_t^{i_{type}}$ represents the vector comprising water-level data or rainfall data at each moment of node i within the specified period t_k ; H_{type} comprises multiple node vectors $h_t^{i_{type}}$, forming the node's level or rainfall attribute matrix.

2.2. Construction of spatiotemporal graph convolutional networks

To effectively correlate the spatial dependency and temporal dynamics, the STGCN model was proposed to combine $\bar{A} \in R^{n \times g}$ and $T \in R^{n \times k \times 2}$ to simultaneously extract the spatiotemporal features of various nodes within the UDN. Fig. 1 shows the overall workflow. First, a temporal convolutional module is employed to extract the temporal features from $T \in R^{n \times k \times 2}$. Subsequently, in the graph convolutional module, $\bar{A} \in R^{n \times g}$ is used to perform spatial convolution on the temporal features of the node information extracted by the temporal convolution module. This process facilitates the aggregation and propagation of information between the nodes within the UDN. Finally, the integrated

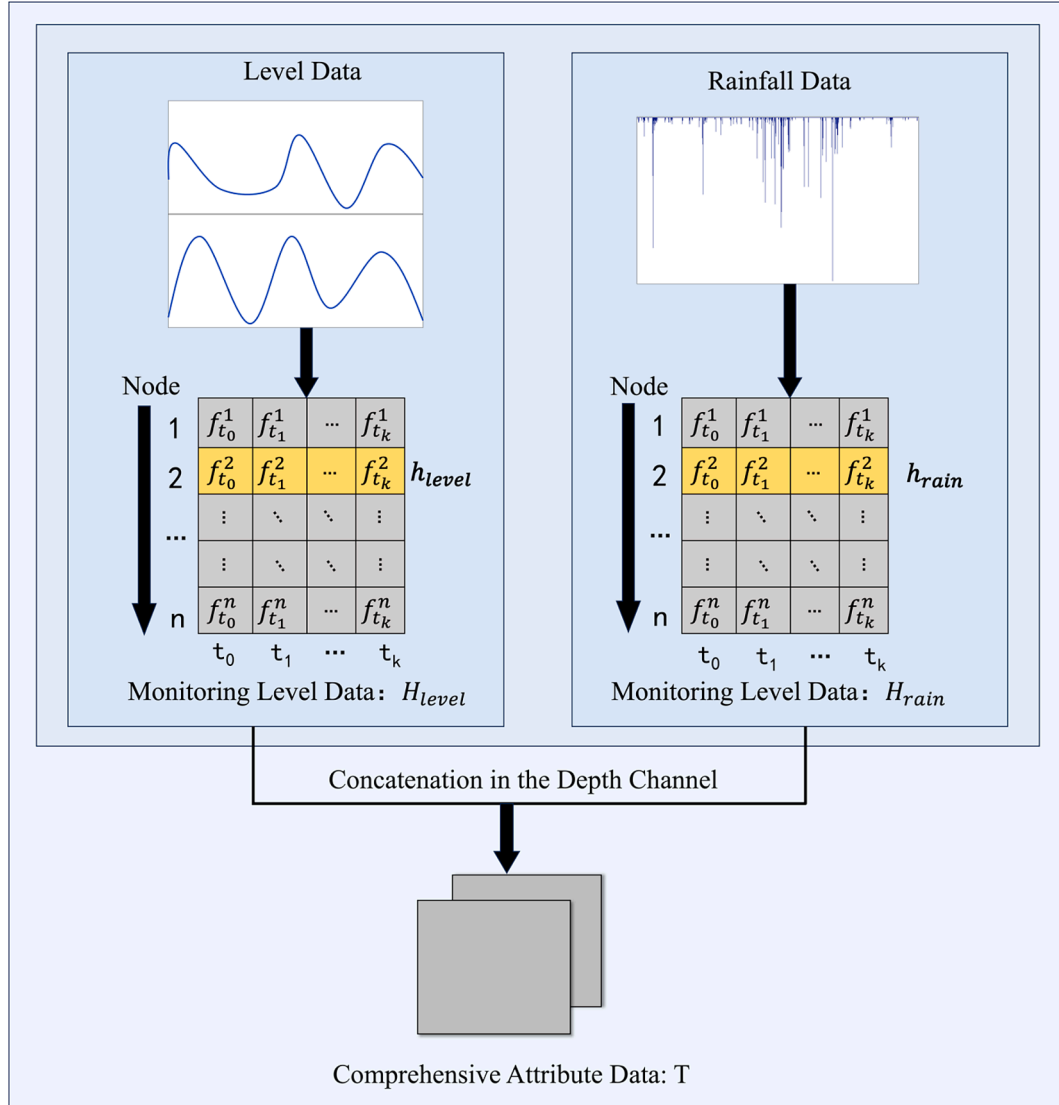


Fig. 3. The processing of temporal dynamic data (water levels and rainfall conditions within nodes).

spatiotemporal features undergo nonlinear transformation through a fully connected layer, leading to the generation of water-level data outputs of the UDNs.

$$r_i^{(z)} = \sum_{i \in V} \bar{A}_i T_i^{(z)} \quad (8)$$

where z denotes the layer within the neighborhood computational graph; $\bar{A}_i$ is the low-dimensional continuous dense matrix after the dimensionality reduction of the high-dimensional sparse matrix $\tilde{A}$; T_i represents the comprehensive attributes of node i , including water-level and rainfall data.

2.3. Passive masked training strategy

An innovative model training strategy, namely the passive-masked training strategy (PMTS), was introduced in this study. Considering a network of n nodes, where q nodes are equipped with water-level monitoring sensors, q_a nodes are selected from q as the input dataset N_a ; q_b nodes are chosen from q for the loss calculation dataset N_b during the model training process (where $q_b = q - q_a$, $N_a \cap N_b = \emptyset$). The unmonitored nodes, denoted by N_c , serve as the test datasets to evaluate the generalizability of the model across the unmonitored nodes. To assess the influence of sensor quantity on model performance, a two-cycle model training strategy was designed, where the outer loop was responsible for adjusting the number of sensors incorporated into the model, while the inner loop was dedicated to the selection of sensor locations, determination of the basis for loss calculation, and initialization of model network parameters (Fig. 4). P_{outer} and P_{inner} represent the specified numbers of outer-loop and inner-loop cycles during training, respectively. Upon model training completion, the

generalizability of the model at the unmonitored nodes was evaluated by calculating the mean absolute error (MAE) and mean absolute percentage error (MAPE) between the real and reconstructed water-level values at the unmonitored nodes on the test datasets.

2.4. Evaluation indicators

2.4.1. Training and validation loss

The mean-squared error (MSE) is employed as the criterion during the training process because of its sensitivity to large errors. By penalizing larger deviations more heavily through squaring the errors between the real and reconstructed values, MSE ensures the model minimizes significant discrepancies and achieves greater robustness. The training and validation loss are calculated using Eq. (9):

$$Loss = \frac{1}{N_q} \sum_{i=1}^{N_q} (L_i - \hat{L}_i)^2 \quad (9)$$

where N_q is the number of sensors selected for the loss calculation; L_i is the real water-level value; $\hat{L}_i$ is the reconstructed water-level value.

2.4.2. Reconstruction performance

In this study, MAE and MAPE are jointly utilized to assess the accuracy of the model in reconstructing water-level data at unmonitored nodes, leveraging their complementary strengths. MAE offers an interpretable measure of absolute errors, ensuring practical relevance across diverse data scales, while MAPE quantifies relative errors as percentages, providing a scale-invariant assessment critical for comparative and real-world applications. The calculation formulae are expressed as in Eqs. (10) and (11):

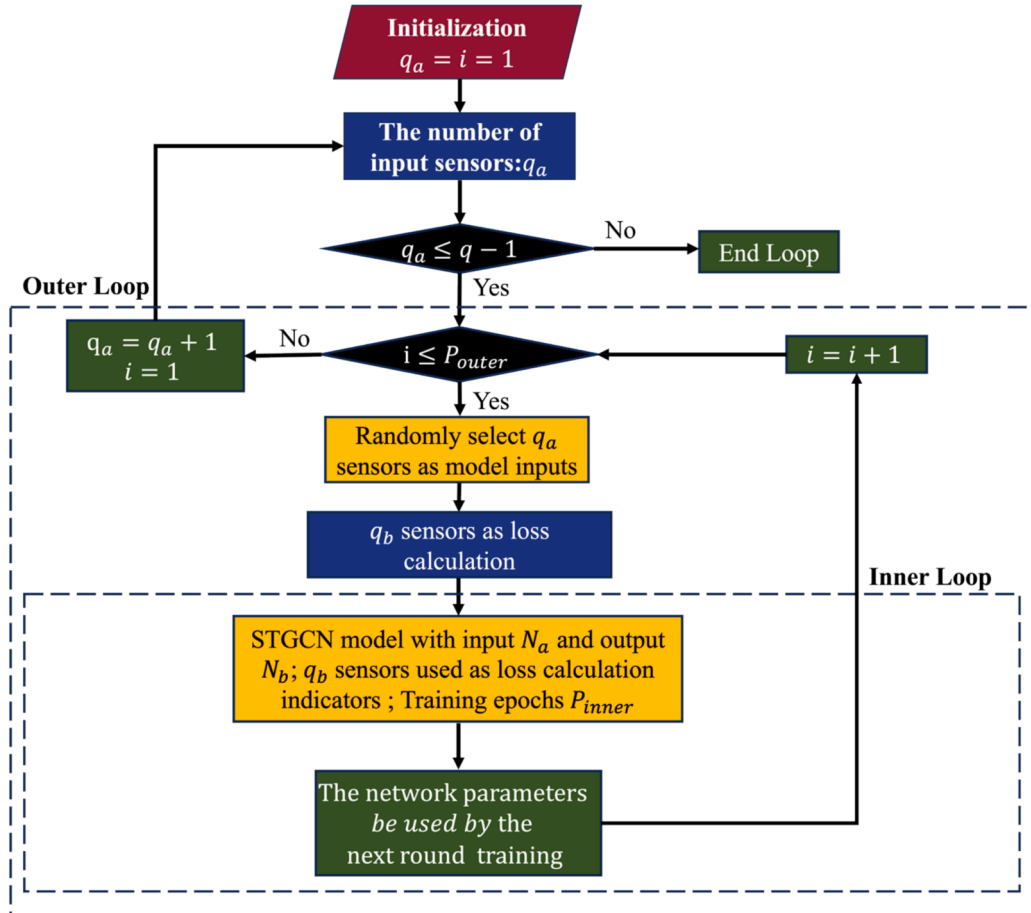


Fig. 4. The flowchart of the passive masked training strategy.

$$MAE = \frac{1}{N_V} \sum_{i=1}^{N_V} |L_i - \hat{L}_i| \quad (10)$$

$$MAPE = \frac{100}{N_V} \sum_{i=1}^{N_V} \left| \frac{L_i - \hat{L}_i}{L_i} \right| \quad (11)$$

where L_i is the real water-level value; $\hat{L}_i$ is the reconstructed water-level value; N_V is the number of nodes with no sensors deployed in the UDN.

3. Case study

The drainage system used in this study is located in Shanghai, China. Fig. 5 shows the network topology with other detailed network information. It is a combined sewer system with a terminal pump, comprising 534 nodes (532 inspection wells, 1 pumping station forebay, and 1 drainage outfall), connected by 539 pipelines. The pumping station, situated at the terminus of the system, has a design flow capacity of 10.72 m³/s. The system covers an area of 172 ha, with a total pipeline length of 12,514 m.

3.1. Spatial static data and temporal dynamic data

The topology diagram of the S drainage system constituted the spatial static data used in this study. For the temporal dynamic data, sensors were deployed at seven nodes within the UDN to monitor water-level fluctuations (Fig. 5), with data collected at 5-min intervals, forming the real-world temporal dataset. Due to the lack of real-world temporal data labels, a simulation model of the system was constructed using InfoWorks ICM (integrated catchment management for urban drainage systems), which has been used to simulate the entire hydrological process of rainfall events by utilizing rainfall, runoff, routing, and pipe flow modules (Elliott & Trowsdale, 2007), to evaluate the accuracy of the STGCN model in reconstructing data at unmonitored nodes. The simulation model was driven by actual rainfall data from June 1, 2020, to June 30, 2020, recorded at 5-min intervals, yielding 8,640 data

points. The simulated water-level data were outputted at the same 5-min intervals, generating an equivalent dataset of 8,640 data points per node. This comprehensive dataset enables a thorough evaluation of the dynamic behavior of the system under various hydrological conditions.

3.2. STGCN model application

In this case study, the network topology was represented by the topology diagram $G \in (534, 539)$, generating the adjacency matrix $A \in R^{534 \times 534}$. The node attribute data $T \in R^{534 \times 8640 \times 2}$ comprised rainfall data H_{rain} and water-level data H_{level} . Employing a time-sliding window approach with a 1-h (12 data points) window width and a stride of 1, a comprehensive node attribute dataset $T \in R^{534 \times 8640 \times 2}$ was segmented, yielding 8,629 data samples. These samples were then partitioned into training, validation, and test sets with a split ratio of 7:1:2.

Table 1 presents the optimal parameter settings for constructing the STGCN model, including the number of hidden layers, number of neurons per layer, learning rate, batch size, and activation function. Table 2 presents the hyper-parameter search range and optimized configurations for each module of the STGCN model. All the experiments were conducted on a GeForce RTX 4090 GPU, 16348CUDA 24 GB Turbo fan 2-slot, and AMD EPYC 7763/64Core/128Threads/2.45G/280 W/256 MB. During the training process, a specified subset of sensors from n_1 to n_7 was randomly selected as the model input, while the unselected sensors were used for loss computation. Upon the completion of model training, the generalizability was evaluated using the synthetic water-level data obtained from the simulation model at all the unmonitored nodes.

4. Results and discussion

The primary application scenario of this study is a real-world situation where data collection in a UDN is extremely limited. Therefore, the key challenge is to fully utilize the available information to accurately reconstruct the water-level data at unmonitored nodes, ensuring data availability for engineering applications. Given the complex hydraulic

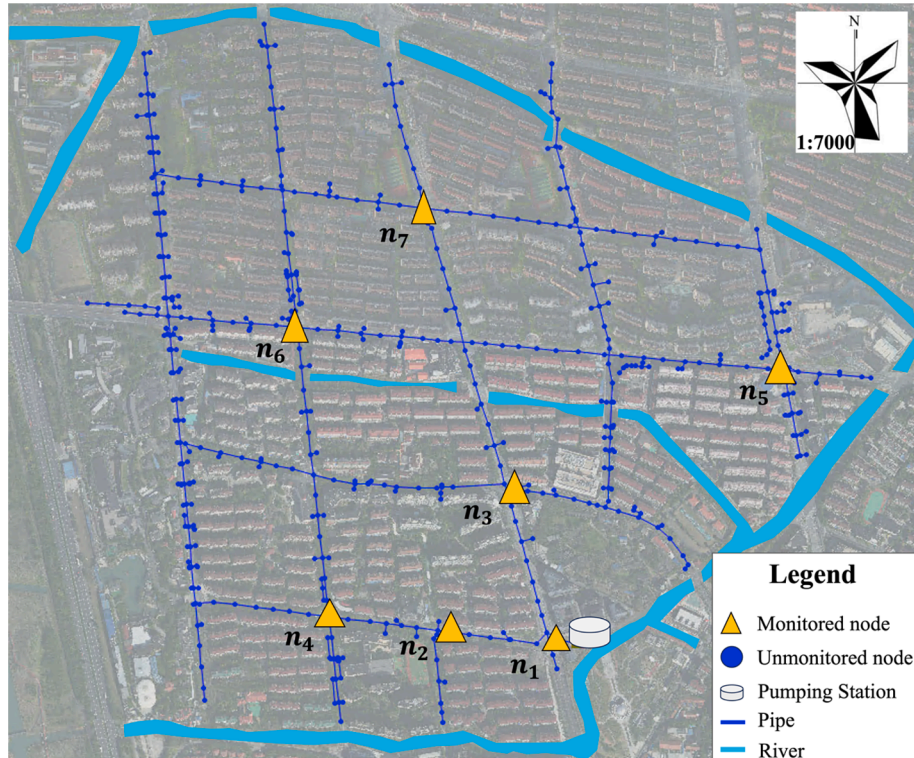


Fig. 5. Topological graph of S drainage system with detailed drainage network information.

Table 1

Model structure and model parameters.

Parameter	TimeBlock	–	STGCNBlock		–	STGCN		
	Conv1D		TimeBlock	GCN		STGCNBlock	TimeBlock	Fully Connected layer
Number of layers	3		2	1		4	1	1
Number of units	256,128,64		/	/		/	/	1
Learning rate	0.0001		0.0001			0.0001		
Optimizer	Adam		Adam			Adam		
Batch size	64		64			64		
Kernel size	3		/	/		/	/	/

Table 2

The hyper-parameter search range and optimized configurations for each module of the STGCN model.

Model backbone	Model	Hyper- parameter	Search range	Optimal configuration
TimeBlock	Conv1D	Learning rate	[0.01,0.0001]	0.0001
		Bathc size	[16,32,64,128]	64
		Training epoch	[100,200,500,1000]	200
		Number of layers	[1,3,5,7]	3
		Number of units	[32,64,128,256]	[256,128,64]
		Kernel size	[3,5,7]	3
STGCNBlock	TimeBlock	Number of layers	[1,2,3]	2
	GCN	Number of layers	[1,3,5]	1
STGCN	STGCNblock	Number of units	[8,16,32]	16
		Number of layers	[1,2,4,6,8]	4
	TimeBlock	Number of layers	[1,2,3]	1
	Fully Connected layer	Number of layers	[1,2,3]	1
		Number of units	[[1],[64,1],[128,64,1]]	1

correlations between the interconnected nodes in a UDN, this study carefully designed and trained an STGCN model by incorporating spatial location information at the monitoring nodes, temporal hydraulic changes, and topological connections of the network. This allows for a precise reconstruction of water levels at unmonitored nodes, even when information is highly limited.

4.1. Impact of sensor configuration and various training strategies on model performanc

Due to limited monitoring data, to avoid overfitting during the training process of the STGCN model and reduce complex co-adaptation between neurons, this study employed a training method where a randomly selected subset of monitoring nodes is inputted during each epoch, rather than using all the monitoring nodes. In this section, the impact of various training strategies and sensor configurations on the

accuracy of the STGCN in reconstructing water-level data at all the unmonitored nodes is investigated.

Sensors were utilized for model training and loss calculation at ratios ranging from 1:6 to 6:1 to assess the adaptability of different training strategies to sensor variability. Two training strategies, namely masked training strategy (MTS) and PMTS, were employed during model training, along with a traditional training strategy (TTS) as the control. For the MTS, a random sensor combination was selected from specified sensor configuration ratios in each iteration to train for 200 epochs. For the PMTS, a random sensor combination was drawn from specified sensor configuration ratios every 20 iterations (the reason for choosing 20 iterations is explained in Section 4.2) throughout the 200 epochs. For the TTS, a sensor combination was randomly selected from the specified sensor configuration ratio and trained repeatedly for 200 epochs. Each strategy was executed 10 times to ensure stable and reliable experimental outcomes. The MAE across all the unmonitored nodes in the UDN

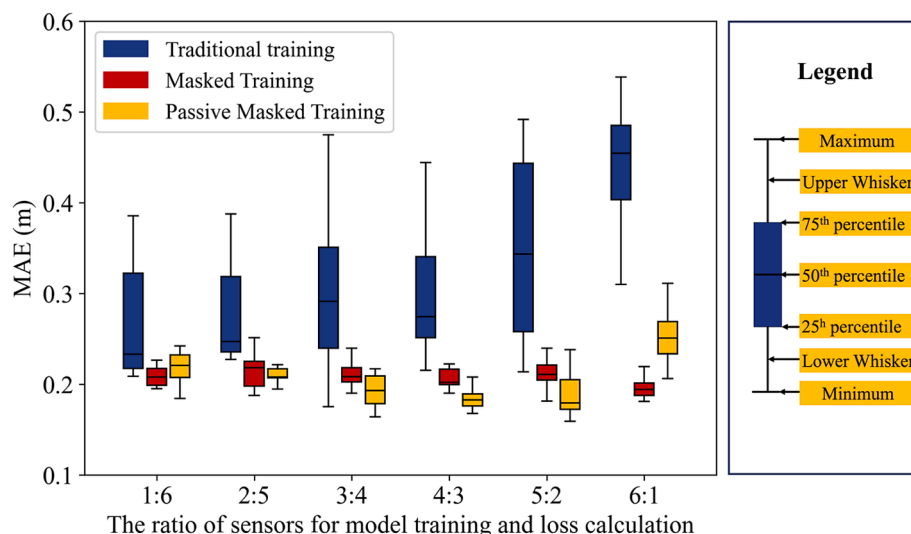


Fig. 6. STGCN model performance at unmonitored nodes under different sensor configuration ratios and different training strategies.

served as the evaluation indicators for model performance.

Fig. 6 compares the MAE results of unmonitored nodes under different training strategies and sensor configuration ratios. The employment of MTS or PMTS significantly enhanced the model's reconstruction accuracy at the unmonitored nodes compared to TTS. Within the TTS framework, the model could only leverage a single sensor combination dataset across different sensor configuration ratios, significantly reducing data diversity and limiting the learning potential and generalizability of the model at the unmonitored nodes. The commonality between the MTS and PMTS is that the positions of the monitoring nodes inputted throughout the training process vary. This training method forces each hidden unit in the STGCN model to learn to work with the randomly selected samples from other units, rather than overly relying on specific units. This approach helps each node to explore its correlation with the monitoring nodes. Under the MTS framework, the MAE values of the model at the unmonitored nodes were 0.208, 0.218, 0.209, 0.202, 0.211, and 0.201 corresponding to sensor configuration ratios of 1:6, 2:5, 3:4, 4:3, 5:2, and 6:1, respectively. With an increase in the number of sensors used for loss computation in the sensor configuration, the variation in the MAE remained modest, ranging from 3 % to 4 %, indicating that the accuracy of water-level data reconstruction at the unmonitored nodes in the UDN had minimal correlation with the sensor configuration under the MTS framework.

In contrast, the PMTS framework demonstrated a significant impact of sensor configuration ratio on the reconstruction performance. The median MAE decreased from 0.221 to 0.179 as the sensor configuration ratio transitioned from 1:6 to 5:2, followed by a substantial increase to 0.251 when the ratio further increased to 6:1. This trend of initial decline and subsequent rise in MAE values, ranging from 13 % to 22 %, highlighted the importance of sensor configuration in the PMTS framework. Iterative training for each sensor combination across different sensor configuration ratios within the prescribed 200 training epochs significantly enhances the spatiotemporal memory capabilities of the STGCN across various nodes, ensuring the capture and retention of unique information from each sensor configuration.

Comprehensive consideration of stability and accuracy revealed that the optimal reconstruction accuracy for the unmonitored nodes was achieved with a sensor configuration ratio of 4:3 under the PMTS framework, despite the lowest median MAE occurring at a ratio of 5:2. Notably, the model's reconstruction accuracy improved by 9 % after iterative training with PMTS compared to MTS. Moreover, a significant finding was that, upon complete training, the water-level data at every drainage network node could be reliably reconstructed using data from only four randomly selected sensor locations. This highlights the pivotal roles of appropriate sensor configuration ratios and PMTS in model architecture and training, offering a novel approach to address data limitations arising from complex and unstable drainage system environments. Unless stated otherwise, subsequent experiments in this study adopted PMTS with a sensor configuration ratio of 4:3, based on the optimal balance between stability and accuracy achieved under these conditions.

4.2. Impact of data dimensionality reduction on model accuracy

To explore the impact of the dimensionality reduction of $\tilde{A}$ on the training performance of the STGCN model, a comprehensive series of experiments with eight different dimensionality reduction ratios (DRRs), ranging from 0.3 to 1.0, was designed. Each experiment was repeated 10 times to ensure the stability and reliability of the results. The effect of the DRR on the data reconstruction performance of the model was analyzed from the loss trend during the training process with the median MAE of all the unmonitored nodes in the test dataset as the evaluation metric.

The validation results obtained during training process demonstrated that the model with dimensionality reduction consistently outperformed

the model without, highlighting the effectiveness of this operation in enhancing training efficiency (Fig. 7). The complexity introduced by the high-dimensional sparse data structure of $\tilde{A}$ might reduce the model's efficiency in extracting relevant information during the training process, which explained the observed improvement in performance with dimensionality reduction. A further analysis of the training process revealed that the model started to learn the information effectively after completing 10 training epochs and achieved a stable accuracy in 20 training epochs. This finding highlights the importance of selecting an appropriate number of training epochs to ensure model accuracy and stability.

Additionally, the impact of different DRR on the model's data reconstruction accuracy was investigated. Initially, increasing the DRR led to a decrease in the median MAE on the test dataset (Fig. 8). However, when the DRR reached 0.6, the median MAE achieved its lowest value (0.204), after which it slightly increased with higher DRR and finally stabilized (0.209–0.211). This suggests that a high DRR could lead to information redundancy, increasing the difficulty of extracting critical information and the computational load, thereby reducing the computational efficiency and increasing the risk of model overfitting. Conversely, a low DRR might result in the loss of critical information, reducing the reconstruction accuracy of the model.

In this study, when the DRR was set to 0.6, the model exhibited the best data reconstruction performance, with the model successfully capturing important spatial information, such as the drainage network topology and rainfall distribution, and effectively reducing the complexity introduced by high-dimensional data. Therefore, unless otherwise specified, the DRR of $\tilde{A}$ was set to 0.6 in all subsequent experiments.

4.3. Impact of historical information input length on model performance

This section examined the influence of historical data input length from monitoring nodes on the STGCN model's data reconstruction accuracy at unmonitored nodes in the UND.

Seven experiments were conducted with input data lengths ranging from 1 h to 8 h (corresponding to historical information lengths of 0 h to 7 h), and unmonitored node data within the current hour as outputs. The performance of the STGCN model was evaluated using the median MAE of the reconstructed water levels at the unmonitored nodes for the current hour, as shown in Fig. 9. Evidently, the incorporation of historical information enhanced the accuracy of the model in reconstructing the water levels at the unmonitored nodes, attributable to the additional information features provided. A comparison of the MAE

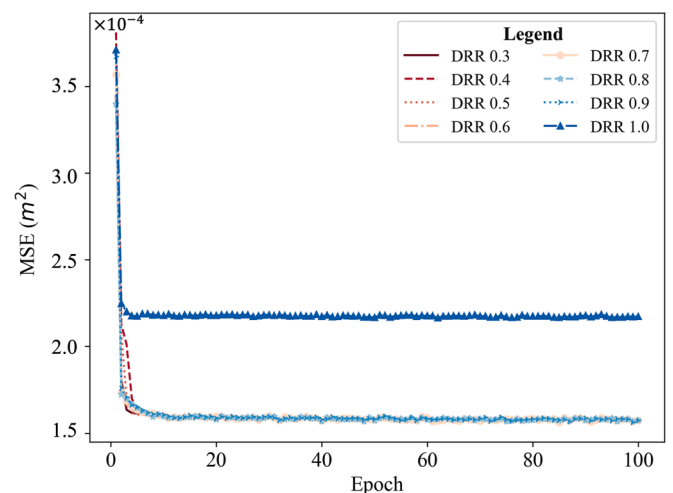


Fig. 7. The validation loss trend of the STGCN model under different dimensionality reduction ratios during the training process.

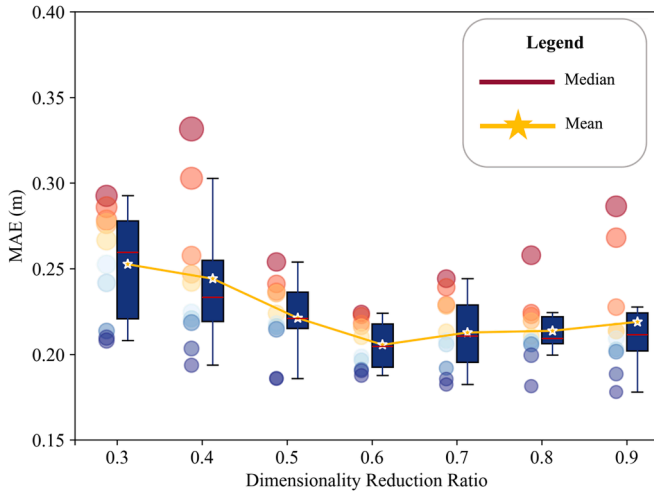


Fig. 8. STGCN model performance at unmonitored nodes in the test dataset under various dimensionality reduction ratios.

values across historical data spanning the time horizon 0 to 7 on an hourly scale revealed a nonlinear relationship between the input length of historical data and reconstruction accuracy. With an increase in the historical information input length, the median MAE initially decreased, reaching a minimum of 0.162 for a historical information input length of 3 h (corresponding to input data lengths of 4 h). However, a further increase in the historical information input length led to a gradual rise in the median MAE, suggesting that beyond a certain threshold, redundant information might “confuse” the capability of the model to extract critical features, thereby reducing the reconstruction accuracy. This may be attributed to the fact that the hydraulic changes at the nodes in a UDN are influenced by upstream inflows and its own previous water levels.

Theoretically, historical data contain information about the hydraulic changes at monitoring nodes over past periods, and more amount of historical data should make it easier to extract hydraulic patterns. Typically, recent historical data have a greater impact on reconstruction, while distant historical data provide less useful information. Therefore, excessively long historical data inputs may lead to increased redundancy, which could mislead the model while extracting key features. Consequently, striking a balance between historical information input length and model reconstruction performance is crucial during the training process. In this study, an optimal historical

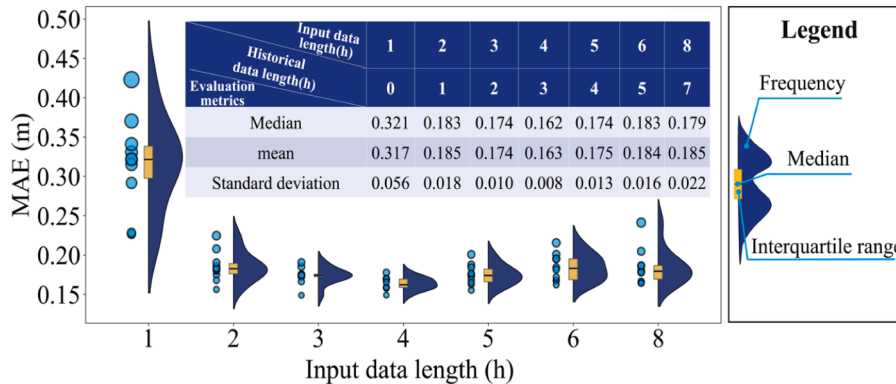
information input length of 3 h (corresponding to input data lengths of 4 h) yielded the highest data reconstruction accuracy. This may be related to the response time of the rainfall-runoff process. In urban or natural hydrological systems, the runoff response triggered by rainfall typically requires a certain amount of time to propagate to downstream monitoring nodes. A 4-hour time window (corresponding to 4-hour input data lengths) may represent the critical response time for the rainfall-runoff process, effectively capturing the primary influence of rainfall on water-level variations at unmonitored nodes. Furthermore, the 4-hour time window might sufficiently encapsulate the key characteristics of these nonlinear relationships between rainfall and water-level changes, providing optimal input features for the spatiotemporal graph convolutional network and facilitating higher reconstruction accuracy.

Note that 1-h input data length denoted the utilization of level monitoring data from known nodes during the current hour to reconstruct water levels at all unmonitored nodes within the network for the same period, without utilizing historical information.

4.4. Spatial distribution of node reconstruction accuracy based on the STGCN model

Fig. 10 shows the spatial distribution of the node reconstruction accuracy across all the unmonitored nodes on the test datasets under the following conditions: a dimensionality reduction ratio of 0.6, a sensor configuration ratio of 4:3, the use of the PMTS framework, and a 4-h historical information input length. Based on the computed MAPE, the node reconstruction accuracy could be categorized into four levels: $\text{MAPE} \leq 10\%$, $10\% < \text{MAPE} \leq 15\%$, $15\% < \text{MAPE} \leq 20\%$, and $20\% < \text{MAPE}$. The results demonstrated that the reconstruction accuracy of the model was influenced by the spatial distribution of the unmonitored nodes. Notably, 71.3 % of these nodes exhibited an MAPE below 10 %, corresponding to a median MAE of 0.038 m. Furthermore, upstream nodes exhibited a superior reconstruction accuracy than downstream nodes, attributable to the coexistence of gravity and pressurized flow states prevalent in combined UDNs. The hydraulic state within pipelines is governed by multiple factors other than the upstream hydraulic conditions, including inherent pipeline characteristics (e.g., diameter) and operational schedules implemented at downstream pumping stations, which may lead to temporary inundation and abrupt hydraulic transitions in the main sewage pipes downstream.

Fig. 11 presents a comparison between the reconstructed and real water-level values for typical nodes across different accuracy ranges. When $\text{MAPE} \leq 10\%$, the reconstructed values showed a high degree of consistency with the true values in terms of the trend, achieving a high



Note that 1-h input data length denoted the utilization of level monitoring data from known nodes during the current hour to reconstruct water levels at all unmonitored nodes within the network for the same period, without utilizing historical information.

Fig. 9. STGCN model performance at unmonitored nodes under different input data lengths.

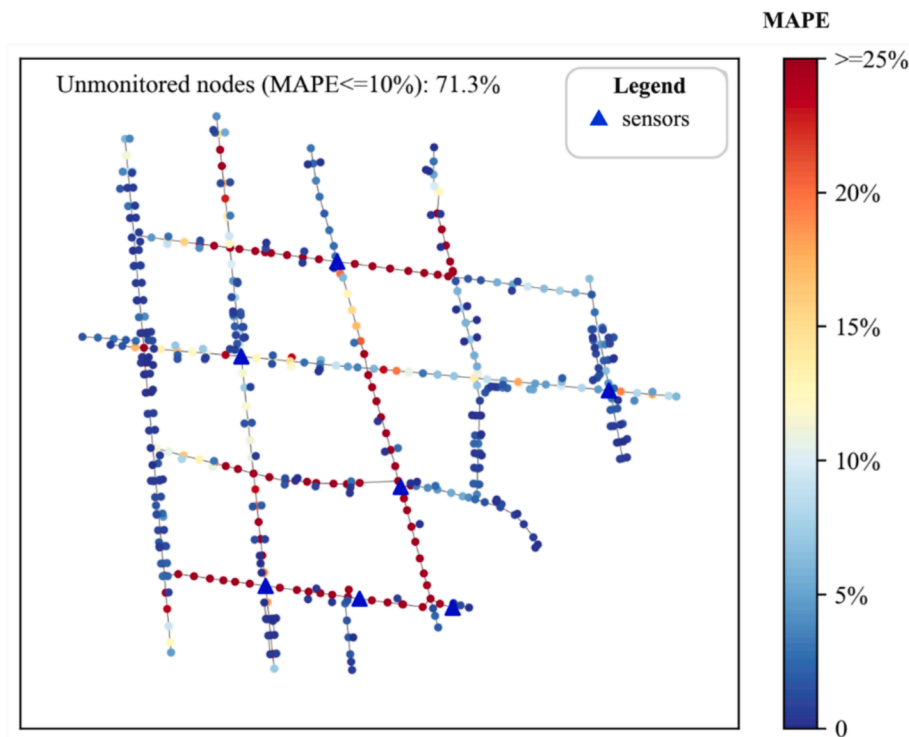


Fig. 10. The spatial distribution of node reconstruction accuracy across all unmonitored nodes.

accuracy in reconstructing the water levels at the unmonitored nodes throughout the entire reconstruction period, with $MAE < 0.1$ m. When $25\% \geq MAPE > 10\%$, the consistency between the reconstructed and true values weakened, leading to noticeable differences during certain periods. Consequently, the accuracy of the reconstructed water levels at the unmonitored nodes decreased over the entire reconstruction period, with $0.1 < MAE < 0.3$ m. These results indicate that in UDNs, the hydraulic states of the interconnected nodes exhibit complex correlations. The hydraulic correlations between monitored and unmonitored nodes can be learned using the STGCN.

This study demonstrated that with limited available data, using only topological structure data, rainfall data, and a small amount of water-level data from a few monitored nodes, the STGCN model, through specific data processing and training strategies, could effectively capture the hydraulic relationships between nodes. The STGCN model achieved a reconstruction accuracy of 71.3 % for the unmonitored nodes in the UDN, proving its practical value in scenarios where monitoring data for drainage networks are sparse. Of course, the incorporation of other physical characteristics, such as the pipe diameter and elevation, may further extend the distribution range of the reconstruction accuracy.

5. Conclusions

In this study, a novel STGCN model was developed to maximize the use of available information to accurately reconstruct the water-level data at unmonitored nodes in UDNs under limited monitoring data. The STGCN model could effectively capture the spatiotemporal relationships between monitored and unmonitored nodes and learn the hydraulic dynamics between nodes. For the first time, the impact of dimensionality reduction operations with varying information retention levels on high-dimensional sparse structural data was investigated considering the loss reduction efficiency during STGCN model training. A novel data construction and training strategy were introduced to enhance the generalizability and reconstruction accuracy of the model at unmonitored nodes. Moreover, the effect of incorporating a dataset of historical information with different lengths during the training process

on the reconstruction accuracy of the model was evaluated. The STGCN model was tested on a real-world drainage network system. The main conclusions of this study are as follows:

- (1) Using PMTS, the STGCN model could more effectively capture the correlations between unmonitored and monitored nodes in the UDN, improving the model's ability to accurately reconstruct water levels at unmonitored nodes.
- (2) Introducing dimensionality reduction on the normalized diffusion matrix ($\tilde{A}$) significantly improved the loss reduction rate during STGCN model training, with an optimal information retention rate of 0.6. This novel approach enhanced data reconstruction accuracy at unmonitored nodes.
- (3) Integrating an appropriate amount of historical water-level data from monitored nodes during training improved the model's understanding of hydraulic characteristics, leading to higher reconstruction accuracy at unmonitored nodes. However, excessive historical data could dilute essential data features, providing insights into optimal data utilization.

Although water-level reconstruction at the unmonitored nodes in UDNs was the primary objective, the capability of the STGCN model to capture spatial and temporal changes in the hydraulic behavior provides a novel approach to address common challenges in the drainage network domain, including limited available monitoring data, substandard data quality, and lack of sensors. For instance, the STGCN model can be employed for the real-time monitoring of anomalous conditions in UDNs and reconstructing information monitoring systems, providing reliable data support for subsequent global optimization and scheduling.

Future studies can focus on improving the accuracy of the STGCN model in reconstructing data at unmonitored nodes along main sewage pipes by involving strongly correlated factors and incorporating additional data, such as the pipeline diameter and operational schedule from terminal pumping stations.

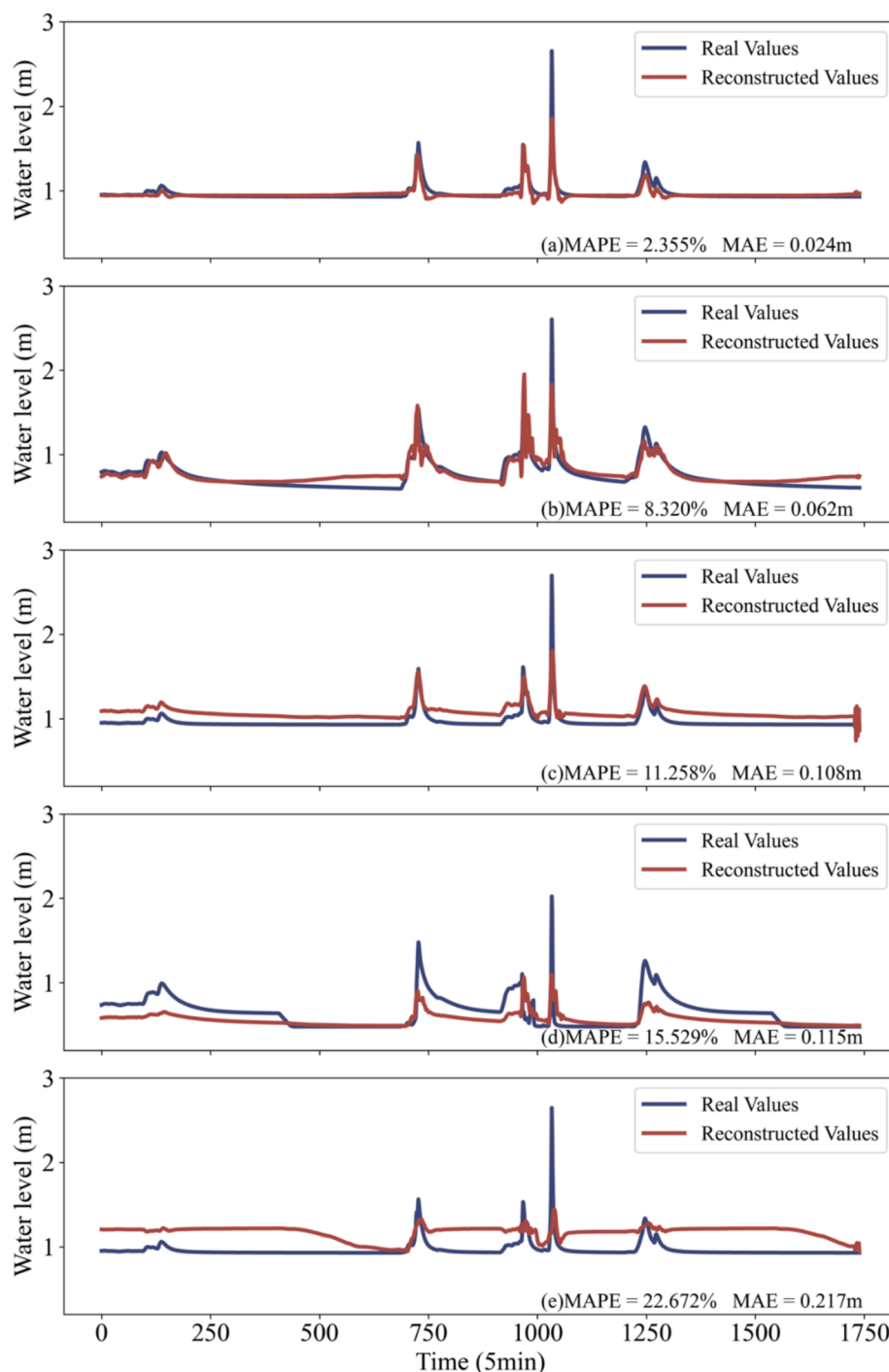


Fig. 11. Examples of the real and reconstructed node water-level values from the experiment results at different accuracy levels(0–10%,10–15%,15–20%,20%–25%).

CRediT authorship contribution statement

Li He: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun Nan:** Supervision, Project administration, Funding acquisition, Conceptualization. **Lei Chen:** Writing – review & editing, Software, Methodology. **Xuesong Ye:** Writing – review & editing, Validation. **Shasha Ji:** Data curation. **Zewei Chen:** Data curation. **Yibo Zhang:** Formal analysis. **Fangmin Wu:** Formal analysis. **Bohan Liu:** Validation. **Zhencheng Ge:** Visualization. **Yanhan Che:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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